

Course Name: Cryptography and
Cyber Security

Course Code: 23CSE163

Module 1

Introduction to cryptography

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Course Learning Objectives:

The students will be able to

1. Enable to learn the fundamental concepts of cryptography, and steganography and make use of these techniques in computing systems.
2. Choose between symmetric and asymmetric encryption techniques depending on the application.
3. Summarize the policies and laws in cyber security.
4. Analyze the security issues and risks in Computer Networks.
5. Analyze the security issues and risks in software and web.

Course Outcomes:

After completing the course, the students will be able to

22CSE163.1 : Apply the concepts of steganography to hide data.

22CSE163.2 : Apply appropriate private or public key encryption techniques depending on the application.

22CSE163.3 : Summarize the policies and laws in cyber security.

22CSE163.4 : Analyze the security issues and risks in Computer Networks.

22CSE163.5 : Analyze the security issues and risks in software

The Growing Importance of Cybersecurity

- Cyber threats are becoming increasingly **sophisticated, impacting businesses, governments, and individuals.**
- With emerging technologies such as **AI, cloud computing, and IoT**, cybersecurity professionals play a crucial role in **ensuring data integrity and privacy.**
- Cybercrime is expected to cost the global economy **trillions of dollars** in the coming years.
- As **cybercriminals** develop more **advanced techniques**, organizations require skilled professionals to implement **proactive security measures, conduct risk assessments, and respond effectively** to incidents.

Cyber Security Career Trends

- According to the **World Economic Forum's** Future of Jobs Report 2025, **Information Security Analysts** rank among the **top 15 fastest-growing professions** globally through **2030**, while **network and cybersecurity skills** are projected to be the **second fastest-growing** skill category worldwide.
- Emerging technologies, particularly **AI and cloud computing**, create **specialized security needs** that drive demand for new skills.
- The cybersecurity job market offers **3.5 million** unfilled cybersecurity positions globally, creating abundant opportunities for qualified professionals.
- The CyberSN report reveals notable growth in specialized roles like **Cybersecurity Technical Writer** (21.73% growth in 2023), **Reverse Engineer/Malware Analyst** (17.38%), and **Vulnerability/Threat Management Analyst** (14.9%).

Top Cybersecurity Career Trends

1. **The Integration of AI and Automation in Cybersecurity**
2. **Increasing Demand for Cloud Security Specialists**
3. **Adoption of Zero Trust Architecture:** insider threats and unauthorized access
4. **Expansion of Cybersecurity Roles in Healthcare:** Protecting sensitive patient data, securing connected medical devices
5. **Increasing Demand for Regulatory Compliance Experts:** GDPR, CCPA, and HIPAA.

- **Cybersecurity Analyst**
- **Penetration Tester (Ethical Hacker)**
- **Cloud Security Engineer**
- **Chief Information Security Officer (CISO)**
- **Incident Response Specialist**
- **Security Architect**

Cyberstorm 2025

Predicting the Next Wave of Threats

AI & Advanced Threats

- AI-Powered Adaptive Malware
- Deepfake-Enabled Attacks
- Enhanced Social Engineering
- Data Poisoning Attacks

Infrastructure Threats

- Critical Infrastructure Attacks
- Cloud & API Vulnerabilities
- Supply Chain Compromises
- IoT & Edge Device Exploitation

Financial & Identity Threats

- Fake Government Apps
- Investment Platform Fraud
- Cryptojacking Attacks
- Identity Theft Campaigns

Ransomware Evolution

- Double-Extortion Tactics
- Physical Infrastructure Targeting
- OT/IoT System Exploitation
- Supply Chain Ransomware

Mobile & Device Threats

- Advanced Mobile Malware
- Cloud-Controlled Android Threats
- Biometric Data Exploitation
- AR System Attacks

Emerging Tech Vulnerabilities

- Zero-Day Exploits
- Quantum Computing Threats
- Advanced AI System Attacks
- AR/VR Platform Vulnerabilities

Bigger spectrum of detections and endpoints

Over 369.01 million detections recorded across **8.44 million** endpoints

Averaging **702 detections** per minute

Behavioral-based detection

53.73 million behavior-based detections, accounted for **14.56%** of total detections

Significant increase from **12.5%** in 2023

~12 attacks per month per personal end user device

Primary attack vectors

68% of attacks originated from Trojans and Infectors

Cloud detections contributed

62% while 38% detections were on-premise

Malware and ransomware incidents

1 Malware incident per **40,436** detections

1 Ransomware incident per **595** detections

Approximately **1 million** Ransomware detections reported over the year

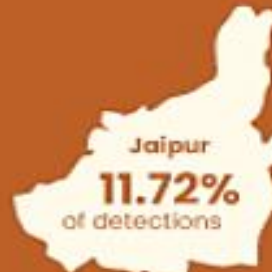
Top most targeted industries

21.82% 
Healthcare

19.57% 
Hospitality

17.38% 
BFSI

Geographical Hotspots



Major Recent Data Breaches Worldwide

- **1. T-Mobile Data Breach (2024):** customer information which included SSN and financial account data.
- **2. PayPal Credential Stuffing Attack:** gain access to user accounts by using credentials obtained from previous breaches.
- **3. LastPass Security Incident:** Attackers breached LastPass, a popular password management service by stealing the encrypted password vaults

Recent Data Breaches in India

- **1. AIIMS Ransomware Attack (2023):** hospital faced major operational disruptions that affected 40 million patient records including their complete medical histories for several weeks.
- **2. HDFC Bank Data Leak:** account numbers and credit card details along with transaction records appeared online
- **3. Indian Railways Data Breach:** exposure of millions of passenger personal data. Travel information together with phone numbers and email addresses
- **4. EdTech Data Leak:** online learning platforms and the protection of student data

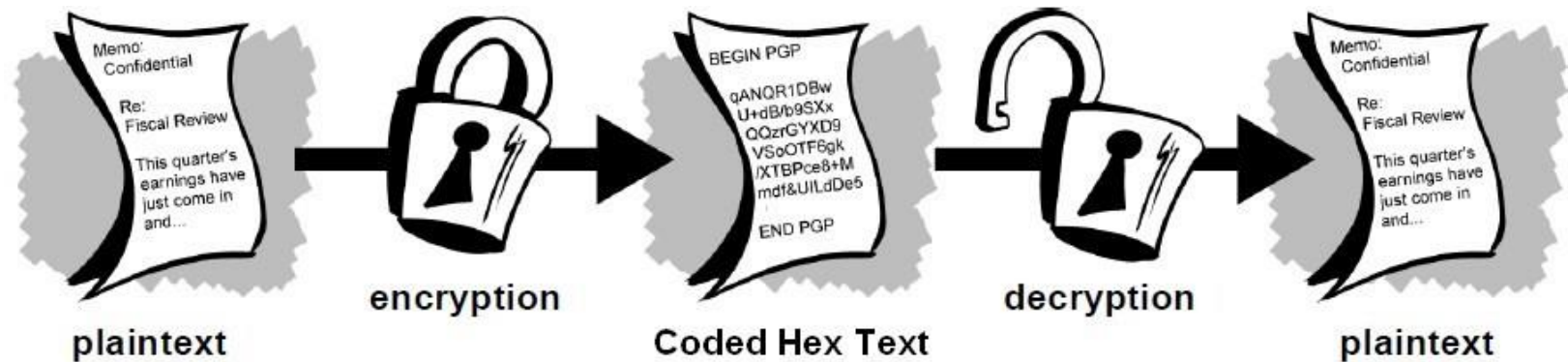
Introduction to Cryptography

- The study of **mathematical techniques for securing digital information, systems, and distributed computations against adversarial attacks.**
- Ex: mechanisms for **ensuring integrity**, techniques for **exchanging secret keys**, protocols for **authenticating** users, electronic **auctions** and **elections**, **digital** cash, and more.
- Classical v/s Modern
- Designing codes- ciphers

Cryptography

- A message is **plaintext** (sometimes called **cleartext**). The process of disguising a message in such a way as to hide its substance is **encryption**. An encrypted message is **ciphertext**. The process of turning ciphertext back into plaintext is **decryption**.

- A **cipher** (or **cypher**) is an algorithm for performing encryption or decryption—a series of well-defined steps that can be followed as a procedure.



Cryptography

- A **cryptosystem** is an implementation of **cryptographic techniques and their accompanying infrastructure** to provide information security services. A cryptosystem is also referred to as a **cipher system**. The various components of a basic cryptosystem are as follows –
 - Plaintext
 - Encryption Algorithm
 - Ciphertext
 - Decryption Algorithm
 - Encryption Key
 - Decryption Key

Cryptography

- While **cryptography** is the science of securing data, **cryptanalysis** is the science of **analyzing and breaking** secure communication.
- Classical cryptanalysis involves an interesting combination of **analytical reasoning**, application of **mathematical tools**, **pattern finding**, patience, determination, and luck.
- **Cryptanalysts** are also called attackers.
- **Cryptology** embraces both cryptography and cryptanalysis.

Cryptography- Goal and Services

- **Goal:** The primary goal of cryptography is to **secure important data** on the hard disk or as it passes through a medium that may not be secure itself. Usually, that medium is a computer network.
- **Services:** Cryptography can provide the following services:
 - ***Confidentiality (secrecy)***
 - ***Integrity (anti-tampering)***
 - ***Authentication***
 - ***Non-repudiation.***



Confidentiality (secrecy)

- Ensuring that no one can read the message except the intended receiver
- Data is kept secret from those without the proper credentials, even if that data travels through an insecure medium



Integrity (anti-tampering)

- Assuring the receiver that the received message has not been altered in any way from the original.



Authentication

- Cryptography can help establish identity for authentication purposes. The process of proving one's identity. The primary forms of host-to-host authentication on the Internet today are name-based or address-based, both of which are notoriously weak.



Non-repudiation

- A mechanism to prove that the sender really sent this message.



Types of Cryptography

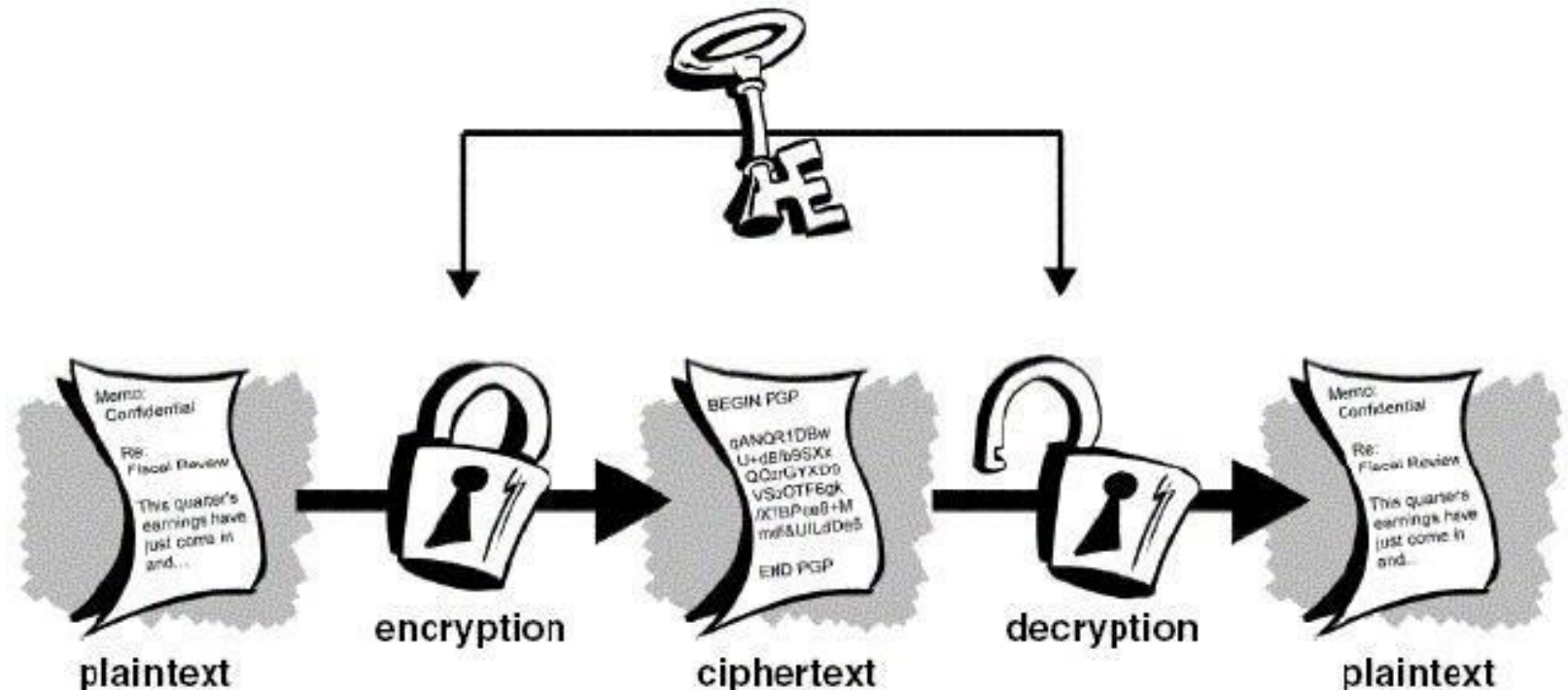
- **Symmetric Key Cryptography**
- **Asymmetric Key Cryptography**
- **Hash Functions**

Symmetric Key Cryptography

- Also known as **Secret Key Cryptography** or Conventional Cryptography, Symmetric Key Cryptography is an encryption system in which the sender and receiver of a message **share a single, common key** that is used to encrypt and decrypt the message.
- The Algorithm use is also known as a **secret key algorithm** or sometimes called a symmetric algorithm
- **A key** is a piece of information (a parameter) that **determines the functional output of a cryptographic algorithm** or cipher.

Symmetric Key Cryptography

- The **key** for encrypting and decrypting the file had to **be known to all the recipients**. Else, the message could not be decrypted by conventional means.
- Example: DES, AES,
- Lucifer, FEAL, LOKI, CAST
- Safer, RC5, Blowfish



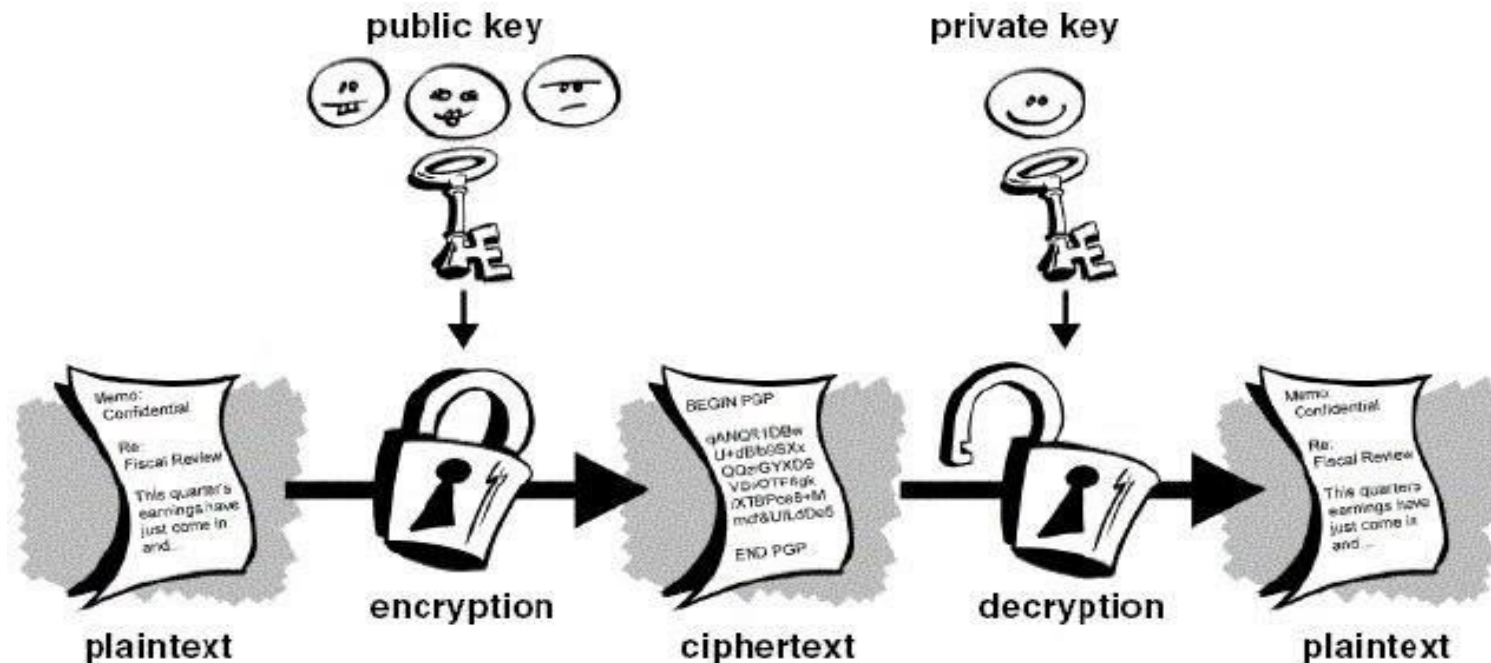
Problems with Conventional Cryptography

Key Management

- Symmetric-key systems are simpler and faster; their main drawback is that the **two parties must somehow exchange the key in a secure way** and keep it secure after that.
- Key Management caused nightmare for the parties using the symmetric key cryptography. This gave the chance for third parties to **intercept the keys in transit** to decode the top-secret messages. Thus, if the **key was compromised**, the entire coding system was compromised and a “Secret” would no longer remain a “Secret”.
- This is why the **“Public Key Cryptography”** came into existence.

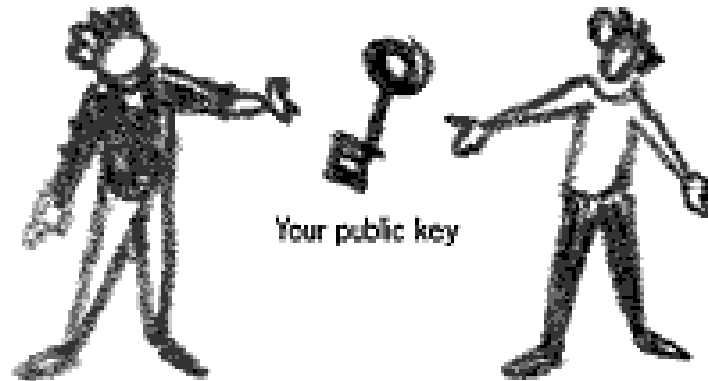
Asymmetric Key Cryptography

- Asymmetric cryptography, also known as **Public-key cryptography**, refers to a cryptographic algorithm which **requires two separate keys**, one of which is **private** and one of which is **public**. The public key is used to encrypt the message and the private one is used to decrypt the message.

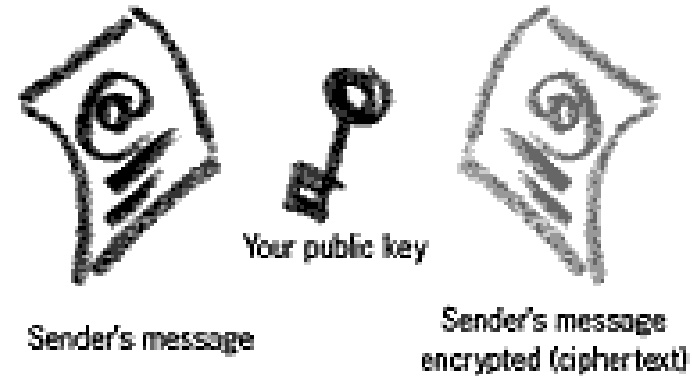


Asymmetric Key Cryptography

Step 1: Give your public key to the sender



Step 2: Sender uses your public key to encrypt the plaintext



Step 3: Sender gives the ciphertext to you



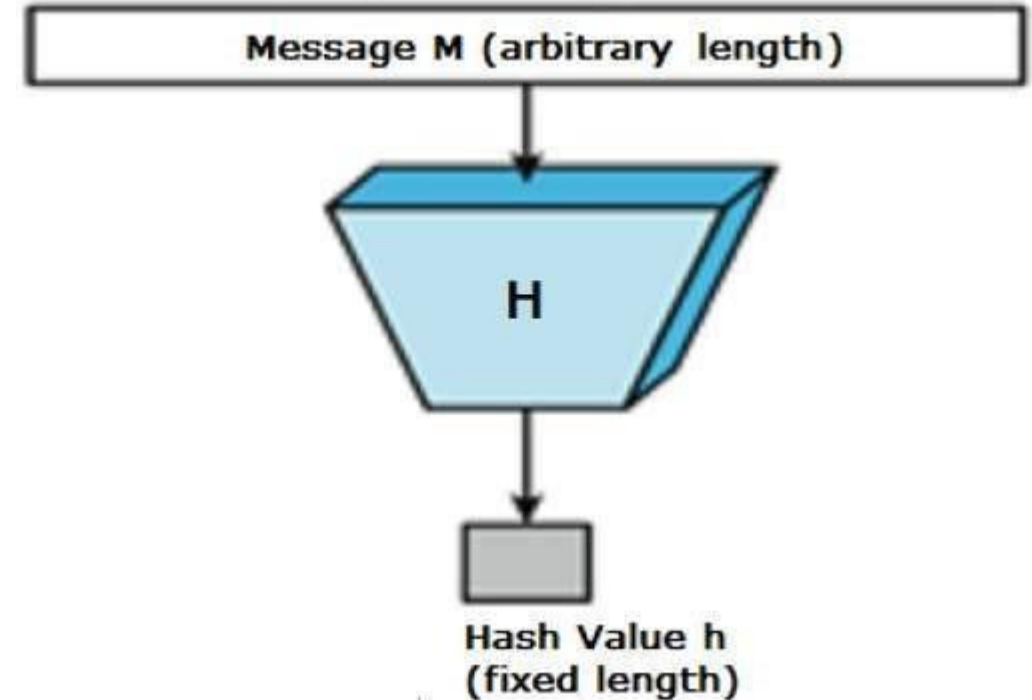
Step 4: Use your private key (and passphrase) to decrypt the ciphertext



Examples: Digital Signature Standard (DSS), RSA (Rivest, Shamir and Adleman) Algorithm

What is a Hash Function?

- A cryptographic hash function is a hash function that **takes an arbitrary block** of data and **returns a fixed-size bit string**, the cryptographic hash value, such that any (accidental or intentional) **change to the data** will (with very high probability) **change the hash** value. The data to be encoded are often called the message, and the hash value is sometimes called the message digest or simply digest.



Terminologies

- An original message is known as the **plaintext**, while the coded message is called the **ciphertext**.
- The process of converting from plaintext to ciphertext is known as **enciphering** or **encryption**; restoring the plaintext from the ciphertext is **deciphering** or **decryption**.
- The many schemes used for encryption constitute the area of study known as **cryptography**.
- Such a scheme is known as a **cryptographic system** or a **cipher**.
- **Techniques used for deciphering a message** without any knowledge of the enciphering details fall into the area of **cryptanalysis**.
- Cryptanalysis - “**breaking the code.**”
- The areas of cryptography and cryptanalysis together are called **cryptology**.

SYMMETRIC CIPHER MODEL

- **Plaintext:** This is the **original intelligible message** or data that is fed into the algorithm as input.
- **Encryption algorithm:** The encryption algorithm **performs various substitutions and transformations** on the plaintext.
- **Secret key:** The secret key is also input to the encryption algorithm. The **key is a value independent of the plaintext** and of the algorithm. The algorithm will produce a different output depending on the specific key being used at the time.
- **Ciphertext:** This is the **scrambled message** produced as output. It depends on the plaintext and the secret key. For a given message, two different keys will produce two different ciphertexts. unintelligible.
- **Decryption algorithm:** This is essentially the **encryption algorithm run in reverse**. It takes the ciphertext and the secret key and produces the original plaintext.

SYMMETRIC CIPHER MODEL

Two requirements:

1. The **opponent** should be **unable to decrypt ciphertext or discover the key**.
2. **Sender and receiver must have obtained copies of the secret key in a secure fashion**

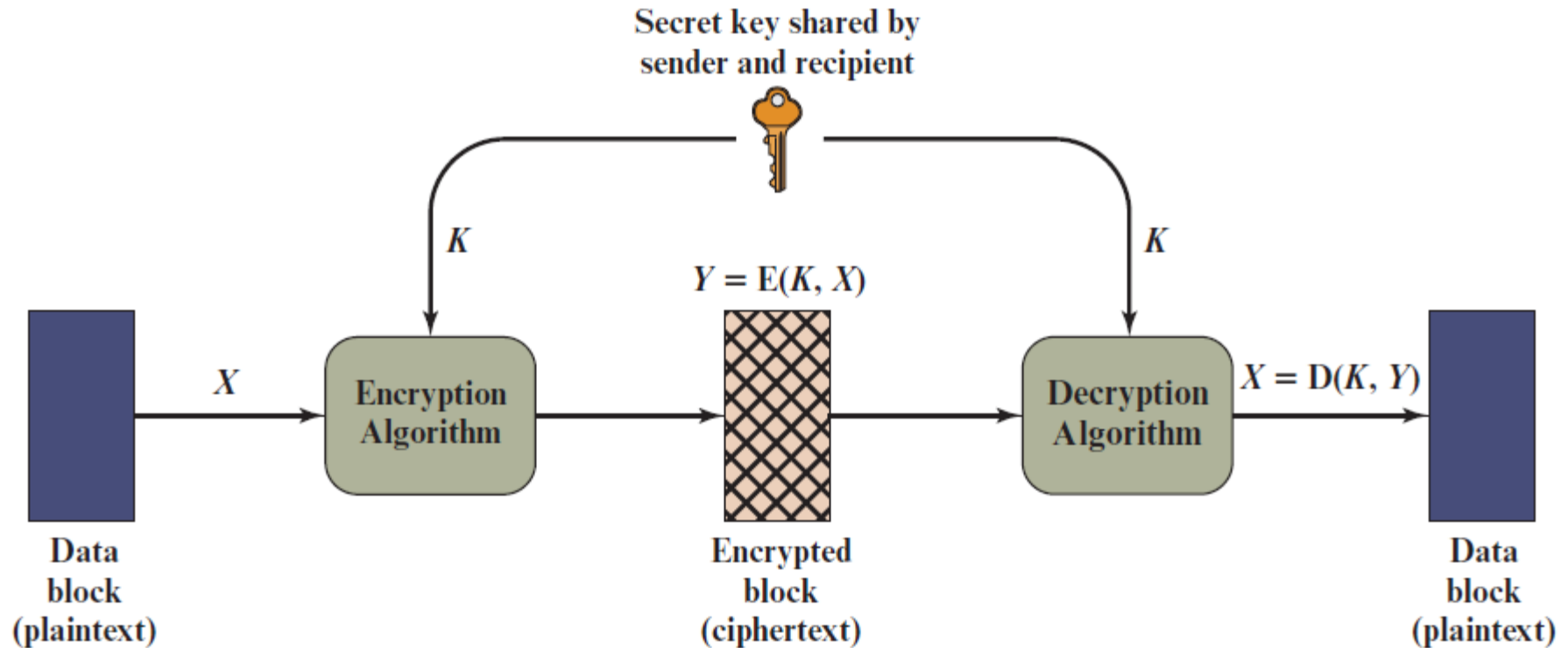


Figure 3.1 Simplified Model of Symmetric Encryption

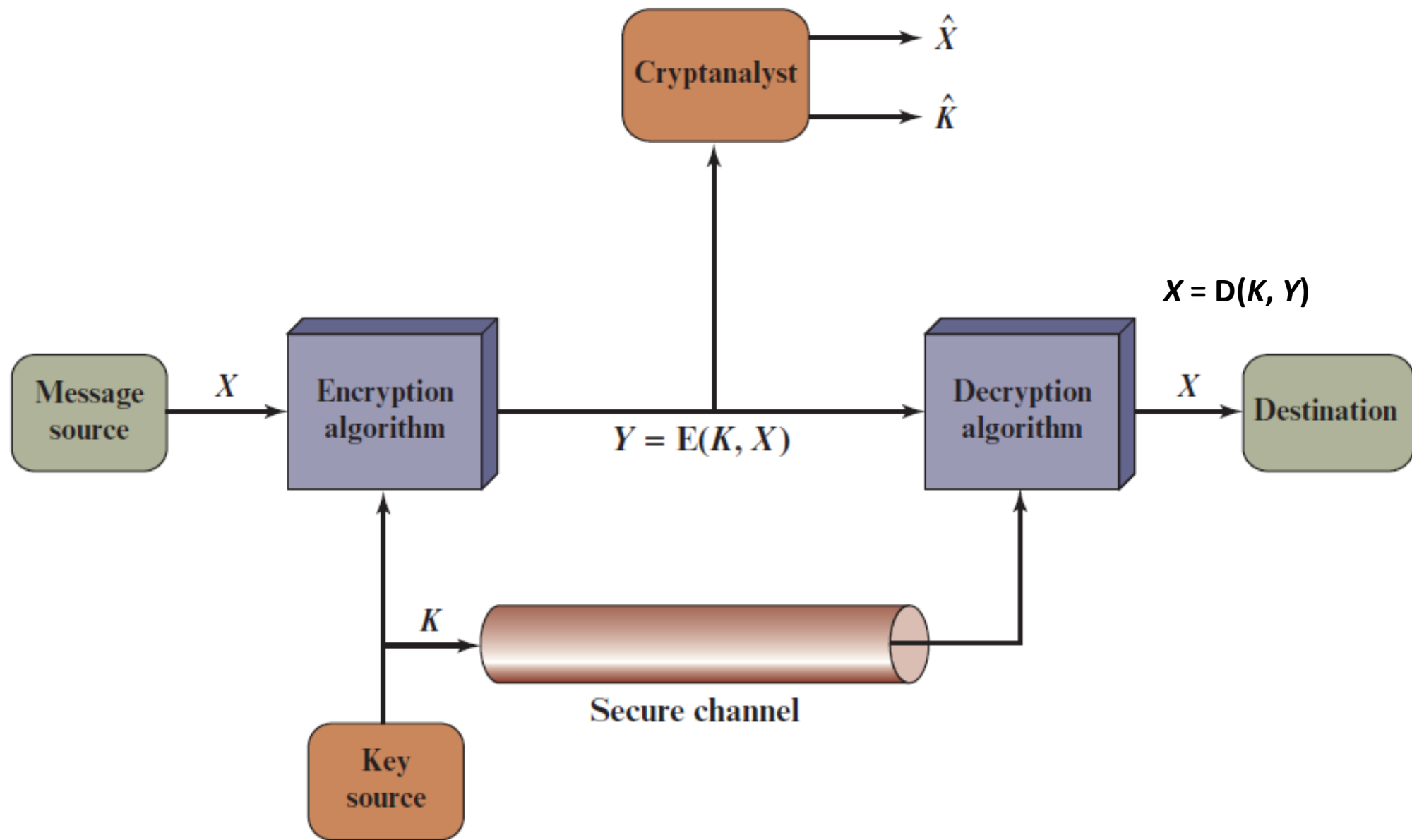


Figure 3.2 Model of Symmetric Cryptosystem

Cryptography

- Cryptographic systems are **characterized** along **three independent dimensions**:
 1. **The type of operations used for transforming plaintext to ciphertext.**
 - **Substitution**, in which each element in the plaintext is mapped into another element
 - **Transposition**, in which elements in the plaintext are rearranged.
 2. **The number of keys used.**
 - **Symmetric**, single-key, secret-key, or conventional encryption.
 - **Asymmetric**, two-key, or public-key encryption.
 3. **The way in which the plaintext is processed.**
 - **Block cipher**
 - **Stream cipher**

Cryptanalysis and Brute-Force Attack

- Typically, the objective of attacking an encryption system is to **recover the key** in use rather than simply to recover the plaintext of a single ciphertext. There are two general approaches to attacking a conventional encryption scheme:

Cryptanalysis: Cryptanalytic attacks **rely on the nature of the algorithm** plus perhaps some **knowledge of** the general characteristics of the **plaintext** or even some sample plaintext–**ciphertext** pairs. This type of attack **exploits the characteristics of the algorithm** to attempt to **deduce a specific plaintext** or to deduce the **key** being used.

Brute-force attack: The attacker **tries every possible key** on a piece of ciphertext until an intelligible translation into plaintext is obtained. On average, **half of all possible keys must be tried** to achieve success.

Types of Attacks on Encrypted Messages

Type of Attack	Known to Cryptanalyst
Ciphertext Only	■ Encryption algorithm ■ Ciphertext
Known Plaintext	■ Encryption algorithm ■ Ciphertext ■ One or more plaintext–ciphertext pairs formed with the secret key
Chosen Plaintext	■ Encryption algorithm ■ Ciphertext ■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	■ Encryption algorithm ■ Ciphertext ■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	■ Encryption algorithm ■ Ciphertext ■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key ■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

SUBSTITUTION TECHNIQUES

- A substitution technique is one in which the **letters of plaintext are replaced by other letters or by numbers or symbols**.
- Replacing plaintext bit patterns with ciphertext bit patterns.

Caesar Cipher

- The Caesar cipher involves **replacing each letter** of the alphabet with the letter standing **three places further down** the alphabet.
- For example,
- plain: meet me after the toga party
- cipher: PHHW PH DIWHU WKH WRJD SDUWB
- Note that the **alphabet is wrapped around**, so that the letter following Z is A.
- We can define the transformation by listing all possibilities, as follows:
- plain: a b c d e f g h i j k l m n o p q r s t u v w x y z
- cipher: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

Caesar Cipher

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12

n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

- For each plaintext letter p , **substitute the ciphertext** letter C :
- $C = E(3, p) = (p + 3) \bmod 26$
- **A shift may be of any amount**, so that the general Caesar algorithm is
- $C = E(k, p) = (p + k) \bmod 26$
- where **k takes on a value in the range 1 to 25**.
- The decryption algorithm is simply
- $p = D(k, C) = (C - k) \bmod 26$
- If it is known that a given ciphertext is a Caesar cipher, then a **brute-force cryptanalysis** is easily performed: simply try all the 25 possible keys.

NOTE: Plaintext is always in lowercase; ciphertext is in uppercase; key values are in italicized lowercase.

Monoalphabetic Ciphers

- With only **25 possible keys**, the Caesar cipher is **far from secure**.
- A dramatic **increase in the key space** can be achieved by allowing an **arbitrary substitution**.
- “cipher” line can be any **permutation of the 26 alphabetic** characters, then there are $26!$ or greater than $4 * 10^{26}$ possible keys.
- Such an approach is referred to as a **monoalphabetic substitution cipher**, because a single cipher alphabet (mapping from plain alphabet to cipher alphabet) is used per message.
- If the cryptanalyst knows the nature of the plaintext (e.g., noncompressed English text), then the analyst can exploit the regularities of the language.

Plain text alphabet:	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Cipher text alphabet (key):	M	U	A	L	V	O	Z	K	R	N	J	X	Q	D	F	S	H	P	E	B	C	T	I

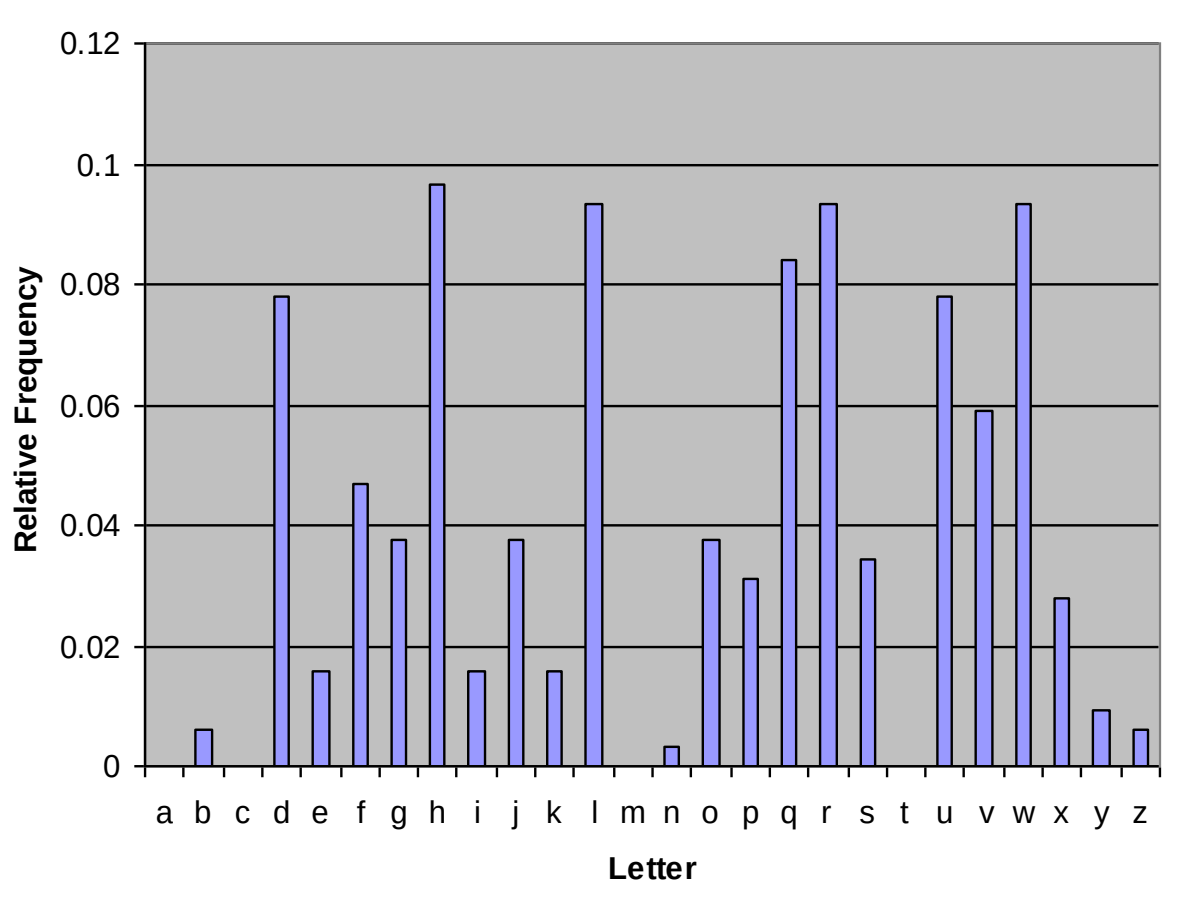
Frequency Analysis

- In English (or any language) **certain letters are used more often than others.**
- If we look at a ciphertext, certain ciphertext letters are going to appear more often than others
- It would be a good **guess that the letters that occur most often** in the ciphertext are actually the most common English letters.
- This is the letter frequency for English.
- The **most common** letter is 'e' by a large margin, followed by 't', 'a', and 'o'
- 'j', 'q', 'x', and 'z' hardly occur at all

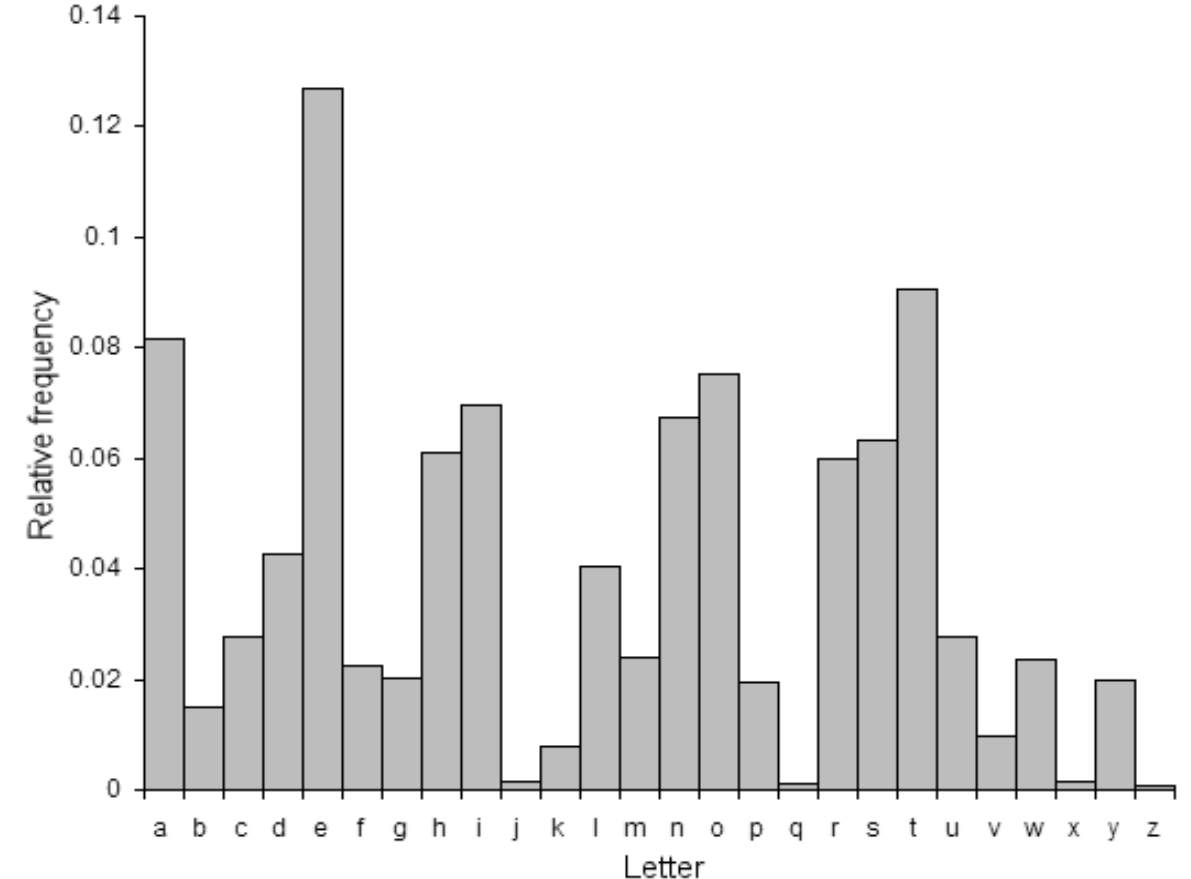
Frequency Analysis in Practice

Suppose this is our ciphertext

dq lqwurgxfwlrq wr frpsxwlqj surylglqj d eurdg vxuyhb ri wkh glvflsolqh dqg dq
lqwurgxfwlrq wr surjudpplqj. vxuyhb wrslfv zloo eh fkrvhq iurp: ruljlqv ri
frpsxwhuv, gdwd uhsuhvhqwdwlrq dqg vwrudjh, errohdq dojheud, gljlwdo
orjlf jdwhv, frpsxwhu dufklwhfwxuh, dvvhpeohuv dqg frpslohuv, rshudwlqj
vbvwHPv, qhwzrunv dqg wkh lqwhuqhw, wkhrulhv ri frpsxwdwlrq, dqg
duwlilfldo lqwhooljhhqfh.



Ciphertext distribution



English distribution

In our ciphertext we have one letter that occurs more often than any other (h), and 6 that occur a good deal more than any others (d, l, q, r, u, and w)

There is a good chance that h corresponds to e, and d, l, q, r, u, and w correspond to the 6 next most common English letters

Frequency Analysis cont.

- If we replace 'e' with 'h' and the 6 next most common letters with their matches, the ciphertext becomes
 - an intro?tion to ?o?p?tin? pro?i?in? a ?roa? ??r?e? o? t?e ?i??ip?ine an? an intro?tion to pro?ra??in?. ??r?e? topi?? ?i?? ?e ??o?en ?ro?: ori?in? o? ?o?p?ter?, ?ata repre?entation an? ?tora?e, ?oo?ean a??e?ra, ?i?ita? ?o?i? ?ate?, ?o?p?ter ar??ite?t?re, a??e???er? an? ?o?pi?er?, operatin? ???te??, net?or?? an? t?e internet, t?eorie? o? ?o?p?tation, an? arti?i?ia? inte??i?en?e.
- A more systematic approach is to **look for other regularities**.
- For example, **certain words may be known to be in the text**. Or we could look for **repeating sequences of cipher letters** and try to deduce their plaintext equivalents.
- A powerful tool is to look at the **frequency of two-letter combinations**, known as **digrams**.

Playfair Cipher

- The best-known **multiple-letter encryption** cipher is the Playfair, which treats **digrams** in the plaintext as **single units** and translates these units into **ciphertext digrams**.
- The Playfair algorithm is based on the use of a **5 x 5 matrix** of letters constructed using a keyword.
- The keyword is *monarchy*.
- The **matrix is constructed** by **filling** in the letters of the keyword (minus duplicates) **from left to right and from top to bottom**, and **then filling in the remainder** of the matrix with the remaining letters in alphabetic order.
- The letters **I and J count as one letter**.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Playfair Cipher

- Plaintext is **encrypted two letters at a time**, according to the following rules:
 1. **Repeating plaintext letters** that are in the same pair are separated with a filler letter, such as x, so that balloon would be treated as ba lx lo on.
 2. Two plaintext letters that fall in the **same row** of the matrix are each **replaced by the letter to the right**, with the first element of the row **circularly** following the last. For example, ar is encrypted as RM.
 3. Two plaintext letters that fall in the **same column** are each replaced by the **letter beneath**, with the top element of the column **circularly** following the last. For example, mu is encrypted as CM.
 4. Otherwise, each plaintext letter in a **pair is replaced by the letter that lies in its own row and the column occupied by the other plaintext letter**. Thus, hs becomes BP and ea becomes IM (or JM)

26 * 26 = 676 digrams - makes frequency analysis much more difficult

Example – Playfair

- Plain Text: "instrumentsz"
- Encrypted text ?

Encrypted Text: gatlmzclrqtx

Encryption:

i -> g, n -> a

s -> t, t -> l

r -> m, u -> z

m -> c, e -> l

n -> r, t -> q

s -> t, z -> x

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Problem: The plaintext is **COMMUNICATE** and the key that we will use to encipher the plaintext is **COMPUTER**. Encrypt using Playfair cipher.

Represents Digraphs Represents Corresponding Cipher

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I
K	L	N	Q	S
V	W	X	Y	Z

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I
K	L	N	Q	S
V	W	X	Y	Z

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I
K	L	N	Q	S
V	W	X	Y	Z

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I
K	L	N	Q	S
V	W	X	Y	Z

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I
K	L	N	Q	S
V	W	X	Y	Z

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I
K	L	N	Q	S
V	W	X	Y	Z

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I
K	L	N	Q	S
V	W	X	Y	Z

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I
K	L	N	Q	S
V	W	X	Y	Z

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I
K	L	N	Q	S
V	W	X	Y	Z

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I
K	L	N	Q	S
V	W	X	Y	Z

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I
K	L	N	Q	S
V	W	X	Y	Z

C	O	M	P	U
T	E	R	A	B
D	F	G	H	I
K	L	N	Q	S
V	W	X	Y	Z

Hill Cipher

- Another interesting **multi-letter cipher** is the Hill cipher, developed by the mathematician Lester Hill in 1929.
- Based on **linear algebra** and **modulo arithmetic**.
- Extendable to work on **different sized blocks** of letters.
- Digraphs, trigraphs ...any sized blocks
- Matrix Multiplication; Modular Inverses; Determinants of Matrices; Matrix Adjugates (for finding inverses).

Encryption - Hill Cipher

- To encrypt a message using the Hill Cipher we must first **turn our keyword into a key matrix** (a **2 x 2 matrix** for working with **digraphs**, a 3 x 3 matrix for working with trigraphs, etc).
- We also turn the **plaintext into digraphs** (or trigraphs) and each of these into a **column vector**. We then perform matrix multiplication modulo the length of the alphabet (i.e. 26) on each vector.
- These vectors are then converted back into letters to produce the ciphertext.
- **2 x 2 Example**
We shall encrypt the **plaintext** message "**short example**" using the **keyword *hill*** and a 2 x 2 matrix. The first step is to turn the keyword into a matrix. If the keyword was longer than the 4 letters needed, we would only take the first 4 letters.
- convert this into a key matrix

$$\begin{pmatrix} H & I \\ L & L \end{pmatrix} \quad \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}$$

Encryption- Hill Cipher

- We now split the plaintext into digraphs, and write these as column vectors.

$$\begin{pmatrix} s \\ h \end{pmatrix} \begin{pmatrix} o \\ r \end{pmatrix} \begin{pmatrix} t \\ e \end{pmatrix} \begin{pmatrix} x \\ a \end{pmatrix} \begin{pmatrix} m \\ p \end{pmatrix} \begin{pmatrix} l \\ e \end{pmatrix} \quad \begin{pmatrix} 18 \\ 7 \end{pmatrix} \begin{pmatrix} 14 \\ 17 \end{pmatrix} \begin{pmatrix} 19 \\ 4 \end{pmatrix} \begin{pmatrix} 23 \\ 0 \end{pmatrix} \begin{pmatrix} 12 \\ 15 \end{pmatrix} \begin{pmatrix} 11 \\ 4 \end{pmatrix}$$

- We multiply the key matrix by each column vector in turn.
- Take each of these numbers in the resultant column vector modulo 26

$$\begin{pmatrix} H & I \\ L & L \end{pmatrix} \begin{pmatrix} s \\ h \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 18 \\ 7 \end{pmatrix} = \begin{pmatrix} 182 \\ 275 \end{pmatrix} = \begin{pmatrix} 0 \\ 15 \end{pmatrix} \text{ mod } 26 = \begin{pmatrix} A \\ P \end{pmatrix}$$

This gives us a final ciphertext of "APADJ TFTWLFJ".

Decryption - Hill Cipher

- To decrypt a ciphertext encoded using the Hill Cipher, we must **find the inverse matrix**. Once we have the inverse matrix, the **process is the same as encrypting**.
- In general, to find the inverse of the key matrix, we perform the calculation below,

$$K^{-1} = d^{-1} \times adj(K)$$

- where **K** is the key matrix, **d** is the determinant of the key matrix and **$adj(K)$** is the adjugate matrix of K .

Step 1 - Find the Multiplicative Inverse of the Determinant

Once we have found this value, we need to take the **number modulo 26**.

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \quad \begin{vmatrix} 7 & 8 \\ 11 & 11 \end{vmatrix} = 7 \times 11 - 8 \times 11 = -11 = 15 \text{ mod } 26$$

We now have to **find the multiplicative inverse of the determinant working modulo 26**. That is, the **number between 1 and 25 that gives an answer of 1 when we multiply it by the determinant**. So, in this case, we are looking for the number that we need to multiply 15 by to get an answer of 1 modulo 26. There are algorithms to calculate this, but it is often easiest to **use trial and error** to find the inverse. So the multiplicative inverse of the determinant modulo 26 is 7.

$$dd^{-1} = 1 \text{ mod } 26$$

$$15 \times 7 = 105 = 1 \text{ mod } 26$$

$$15 \times x = 1 \text{ mod } 26$$

Step 2 - Find the Adjugate Matrix

- We **swap the top left and bottom right numbers** in the **key matrix**, and **change the sign** of the **top right** and **bottom left** numbers.
- Next, we will need to take each of them **modulo 26**.

$$\text{adj} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$\text{adj} \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} = \begin{pmatrix} 11 & -8 \\ -11 & 7 \end{pmatrix} = \begin{pmatrix} 11 & 18 \\ 15 & 7 \end{pmatrix}$$

Step 3 - Multiply the Multiplicative Inverse of the Determinant by the Adjugate Matrix

- To get the inverse key matrix, we **now multiply the inverse determinant** (that was 7 in our case) from step **1 by each of the elements of the adjugate matrix** from step 2. Then we take each of these answers **modulo 26**.

$$K^{-1} = d^{-1} \times adj(K)$$

$$7 \times \begin{pmatrix} 11 & 18 \\ 15 & 7 \end{pmatrix} = \begin{pmatrix} 77 & 126 \\ 105 & 49 \end{pmatrix} = \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \text{ mod } 26$$

$$\text{if } K = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}, \text{ then } K^{-1} = \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix}$$

Decryption

- Now we have the inverse key matrix, we have to **convert the ciphertext into column vectors** and **multiply the inverse matrix by each column vector** in turn, take the results modulo 26 and convert these back into letters to get the plaintext.
- We get back our plaintext of "short example".

$$\begin{aligned}\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} A \\ P \end{pmatrix} &= \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 0 \\ 15 \end{pmatrix} \\ &= \begin{pmatrix} 25 \times 0 + 22 \times 15 \\ 1 \times 0 + 23 \times 15 \end{pmatrix} \\ &= \begin{pmatrix} 330 \\ 345 \end{pmatrix} \\ &= \begin{pmatrix} 18 \\ 7 \end{pmatrix} \text{ mod } 26 \\ &= \begin{pmatrix} S \\ h \end{pmatrix}\end{aligned}$$

Hill Cipher Example

- Plaintext: “attack” , Key: “cddg”
- $C = K P \bmod 26$
- $P = K^{-1} C \bmod 26$
- $K^{-1} = D^{-1} \text{adj}(K)$

•

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

- Key matrix: $\begin{pmatrix} 2 & 3 \\ 3 & 6 \end{pmatrix}$
-

- Ciphertext: FKMPIO

- $D = |12-9| = 3$

- $D \times D^{-1} = 1 \pmod{26}$

- $3 \times D^{-1} = 1 \pmod{26}$

- $3 \times 9 = 27 \pmod{26} = 1 \pmod{26}$

- $D^{-1} = 9$

Polyalphabetic Ciphers - *Vigenère Cipher*

- Another way to **improve** on the simple monoalphabetic technique is to **use different monoalphabetic substitutions** as one proceeds through the plaintext message.
- Assume a sequence of plaintext letters $P = p_0, p_1, p_2, \dots, p_{n-1}$ and a key consisting of the sequence of letters $K = k_0, k_1, k_2, \dots, k_{m-1}$, where typically $m < n$. The sequence of ciphertext letters $C = C_0, C_1, C_2, \dots, C_{n-1}$ is calculated as follows:

$$\begin{aligned} C &= C_0, C_1, C_2, \dots, C_{n-1} = E(K, P) = E[(k_0, k_1, k_2, \dots, k_{m-1}), (p_0, p_1, p_2, \dots, p_{n-1})] \\ &= (p_0 + k_0) \bmod 26, (p_1 + k_1) \bmod 26, \dots, (p_{m-1} + k_{m-1}) \bmod 26, \\ &\quad (p_m + k_0) \bmod 26, (p_{m+1} + k_1) \bmod 26, \dots, (p_{2m-1} + k_{m-1}) \bmod 26, \dots \end{aligned}$$

- Thus, the first letter of the key is added to the first letter of the plaintext, mod 26

$$C_i = (p_i + k_{i \bmod m}) \bmod 26$$

$$p_i = (C_i - k_{i \bmod m}) \bmod 26$$

- To encrypt a message, a **key** is needed that is **as long as the message**. Usually, the key is a **repeating keyword**. For example, if the keyword is *deceptive*, the message “we are discovered save yourself” is encrypted as
- key: *deceptivedeceptivedeceptive*
- plaintext: weare**red**discover**red**saveyourself
- ciphertext: ZIC**VTW**QNGRZG**VTW**AVZHCQYGLMGJ

key	3	4	2	4	15	19	8	21	4	3	4	2	4	15
plaintext	22	4	0	17	4	3	8	18	2	14	21	4	17	4
ciphertext	25	8	2	21	19	22	16	13	6	17	25	6	21	19

key	19	8	21	4	3	4	2	4	15	19	8	21	4
plaintext	3	18	0	21	4	24	14	20	17	18	4	11	5
ciphertext	22	0	21	25	7	2	16	24	6	11	12	6	9

Repeated sequences VTW at a displacement of 9 and make the assumption that the keyword is either three or nine letters in length.

The strength of this cipher is that **there are multiple ciphertext letters for each plaintext letter**, one for each unique letter of the keyword.

Vernam Cipher

- The ultimate defense against such a cryptanalysis is to choose a **keyword** that is **as long as the plaintext** and has **no statistical relationship** to it.
- Such a system was introduced by an AT&T engineer named Gilbert Vernam in 1918. His system works on binary data (bits) rather than letters.

- $c_i = p_i \text{ xor } k_i$

- where

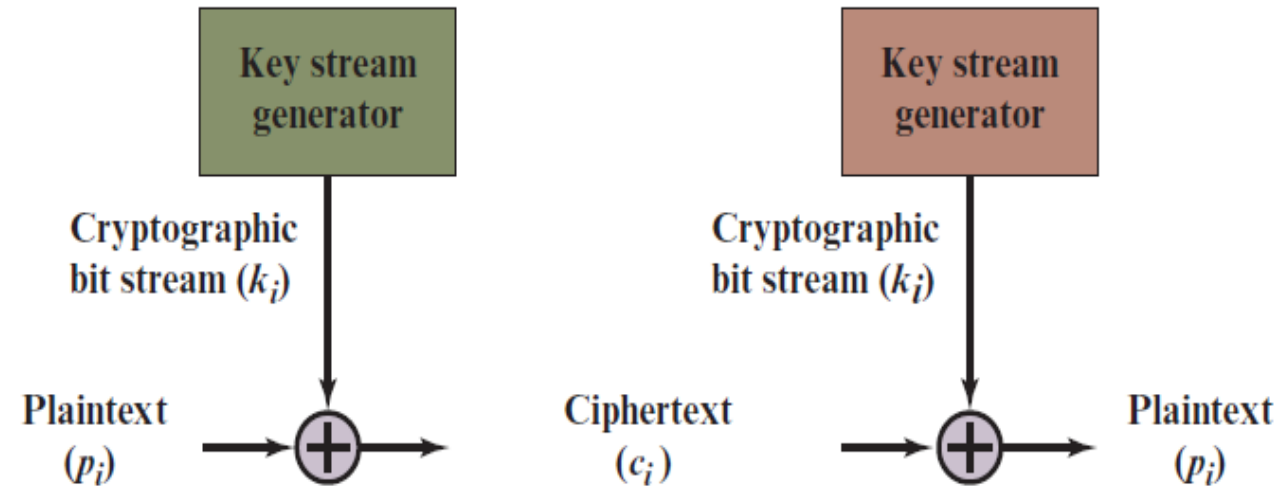
- p_i = i th binary digit of plaintext

- k_i = i th binary digit of key

- c_i = i th binary digit of ciphertext

- Thus, the ciphertext is generated by performing the bitwise XOR of the plaintext and the key.

- $p_i = c_i \text{ xor } k_i$



One-Time Pad

- An Army Signal Corp officer, Joseph Mauborgne, proposed an **improvement to the Vernam cipher** that yields the **ultimate in security**.
- Mauborgne suggested using a **random key that is as long as the message**, so that the key need not be repeated.
- In addition, the **key is to be used to encrypt and decrypt a single message**, and then is **discarded**.
- Each **new message requires a new key of the same length** as the new message. Such a scheme, known as a **one-time pad**.
- Because the **ciphertext contains no information whatsoever about the plaintext**, there is simply **no way to break the code**.

One-Time Pad

ANKYODKYUREPFJBYOJDSPLEIYIUNOFDOIUERFPLUYTS

We now show two different decryptions using two different keys:

ciphertext: ANKYODKYUREPFJBYOJDSPLEIYIUNOFDOIUERFPLUYTS

key: *pxlmvmsydofoyrvzwc tnlebnecvgdupahfzzlmnyih*

plaintext: mr mustard with the candlestick in the hall

ciphertext: ANKYODKYUREPFJBYOJDSPLEIYIUNOFDOIUERFPLUYTS

key: *pftgpmiydgaxgoufhklmhsqdqogtewbqfgyovuhwt*

plaintext: miss scarlet with the knife in the library

- Suppose that a **cryptanalyst** had managed to find these two keys. Two plausible plaintexts are produced.
- There is **no way to decide which key is correct** and therefore **which plaintext is correct**.
- The **security** of the one-time pad is entirely due to the **randomness of the key**.
- If the stream of characters that constitute the key is truly random, then the stream of characters that constitute the ciphertext will be truly random.
- Thus, there are **no patterns** or regularities that a cryptanalyst can use to attack the ciphertext.

One-Time Pad

- The one-time pad offers complete security but, in practice, has two fundamental difficulties:
 1. There is the practical problem of **making large quantities of random keys**. Any heavily used **system might require millions of random characters** on a regular basis. Supplying truly random characters in this volume is a significant task.
 2. Even more daunting is the problem of **key distribution and protection**. For every message to be sent, a key of equal length is needed by both sender and receiver. Thus, a mammoth key distribution problem exists.

TRANSPOSITION TECHNIQUES - Rail fence

- A very different kind of mapping is achieved by performing some sort of **permutation on the plaintext letters**. This technique is referred to as a **transposition cipher**.
- The simplest such cipher is the **rail fence technique**, in which the plaintext is **written down as a sequence of diagonals** and **then read off as a sequence of rows**.
- For example, to encipher the message “meet me after the toga party” with a rail fence of depth 2, we write the following:

```
m e m a t r h t g p r y  
e t e f e t e o a a t
```

- The encrypted message is
- MEMATRHTGPRYETEFETEOAAT

TRANSPOSITION TECHNIQUES

- This sort of thing would be trivial to cryptanalyze.
- A more complex scheme is to **write the message in a rectangle, row by row**, and **read the message off, column by column**, but **permute the order of the columns**.
- The **order of the columns** then becomes the **key** to the algorithm. For example,

• Key: 4 3 1 2 5 6 7

Plaintext: a t t a c k p

o s t p o n e

d u n t i l t

w o a m x y z

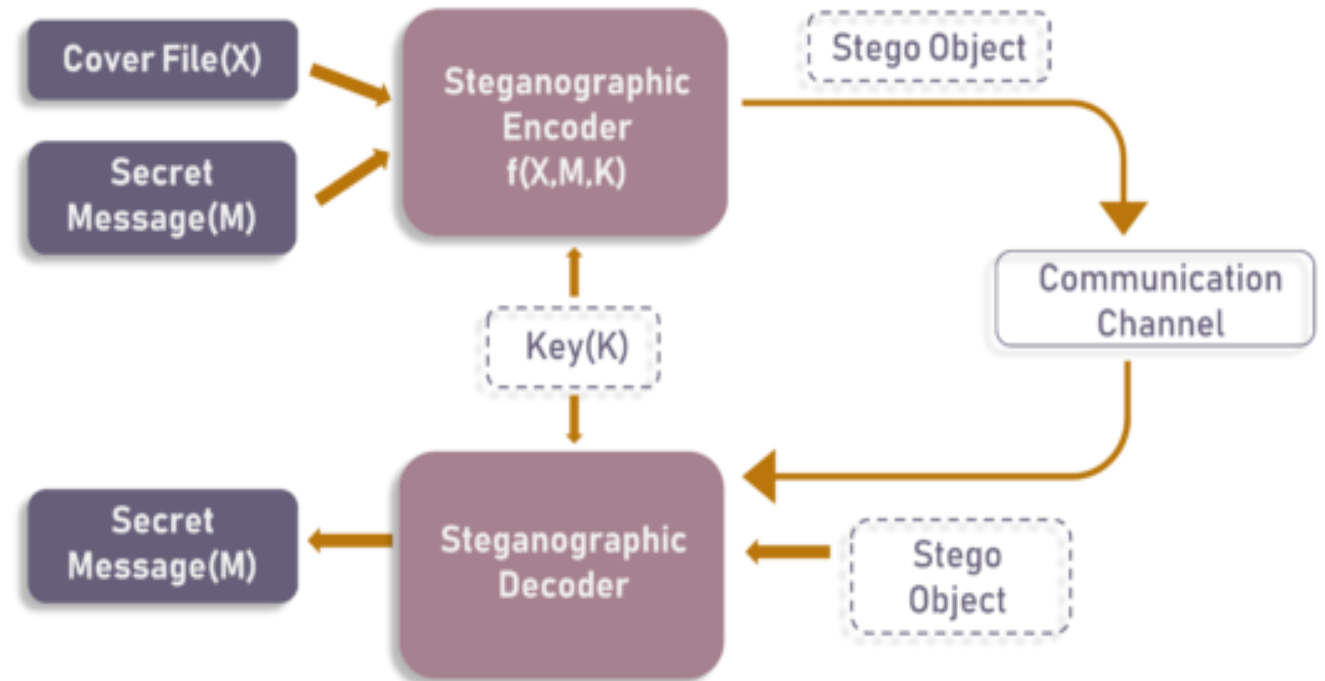
- Ciphertext: TTNAAPTMTSUOAODWCOIXKNLYPETZ
- To encrypt, start with the column that is labeled 1, in this case column 3. Write down all the letters in that column. Proceed to column 4, which is labeled 2, then column 2, then column 1, then Columns 5, 6, and 7.

TRANSPOSITION TECHNIQUES

- **Cryptanalysis** is fairly **straightforward** and involves laying out the ciphertext in a matrix and **playing around with column positions**.
- **Digram and trigram frequency tables** can be useful.
- The transposition cipher **can be made** significantly more **secure** by performing **more than one stage of transposition**. The result is a more complex permutation that is not easily reconstructed.

Steganography

- Steganography is the art and science of **embedding secret messages in a cover message** in such a way that no one, apart from the sender and intended recipient, suspects the existence of the message .
- Resulting Stego Object looks very similar to your cover file, with no visible changes.



	STEGANOGRAPHY	CRYPTOGRAPHY
Definition	It is a technique to hide the existence of communication	It's a technique to convert data into an incomprehensible form
Purpose	Keep communication secure	Provide data protection
Data Visibility	Never	Always
Data Structure	Doesn't alter the overall structure of data	Alters the overall structure of data
Key	Optional, but offers more security if used	Necessary requirement
Failure	Once the presence of a secret message is discovered, anyone can use the secret data	If you possess the decryption key, then you can figure out original message from the ciphertext

Steganography Techniques

- Depending on the **nature of the cover object**(actual object in which secret data is embedded), steganography can be divided into five types:
 - Text Steganography
 - Image Steganography
 - Video Steganography
 - Audio Steganography
 - Network Steganography

Text Steganography

- Text Steganography is hiding information inside the text files. It involves things like **changing the format of existing text, changing words within a text, generating random character sequences or using context-free grammars** to generate readable texts. Various techniques used to hide the data in the text are:
 - Format Based Method
 - Random and Statistical Generation
 - Linguistic Method

Image Steganography

- Hiding the data by taking the cover object as the image is known as image steganography.
- In digital steganography, images are widely used cover source because there are a huge number of bits present in the digital representation of an image. There are a lot of ways to hide information inside an image. Common approaches include:
 - Least Significant Bit Insertion
 - Masking and Filtering
 - Redundant Pattern Encoding
 - Encrypt and Scatter
 - Coding and Cosine Transformation

Audio Steganography

- In audio steganography, the secret message is embedded into an audio signal which alters the binary sequence of the corresponding audio file. Hiding secret messages in digital sound is a much more difficult process when compared to others, such as Image Steganography. Different methods of audio steganography include:
 - Least Significant Bit Encoding
 - Parity Encoding
 - Phase Coding
 - Spread Spectrum
- This method hides the data in WAV, AU, and even MP3 sound files.

Video Steganography

- In Video Steganography you can hide kind of data into digital video format. The advantage of this type is a large amount of data can be hidden inside and the fact that it is a moving stream of images and sounds.
- You can think of this as the combination of Image Steganography and Audio Steganography.
- Two main classes of Video Steganography include:
 - Embedding data in uncompressed raw video and compressing it later
 - Embedding data directly into the compressed data stream

Network Steganography (Protocol Steganography)

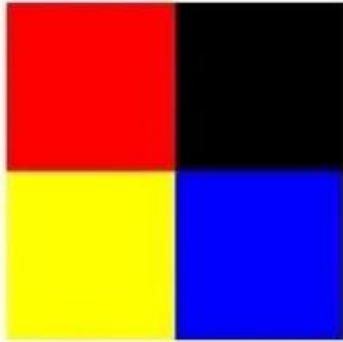
- It is the technique of embedding information within network control protocols used in data transmission such TCP, UDP, ICMP etc.
- You can use steganography in some covert channels that you can find in the OSI model.
- For Example, you can hide information in the header of a TCP/IP packet in some fields that are either optional.

Image Steganography

- Image cover files are a **preferred** choice because **quality degradation** of the image due to encryption **cannot be detected** by the human eye.
- Cover files allow for the encryption of a wide range of data forms including text, images, and multimedia.
- To reduce the size gap between the cover and the hidden image, **least significant bit (LSB)** replacement techniques were introduced. It is possible to change **up to four least significant bits** without making an image significantly different.
- Changing four LSBs of a pixel would mean that the **cover image only needs to be twice the size of the image** that needs to be hidden.
- Obviously, the more of the LSBs being changed means the bigger degradation of image quality as compared to the original.
- Therefore, an **amount of LSBs changed** needs to be such that it has a **reasonable cover** to hidden image size ratio, as well as **maintaining the quality** of the cover image.

Least significant bit technique, LSB.

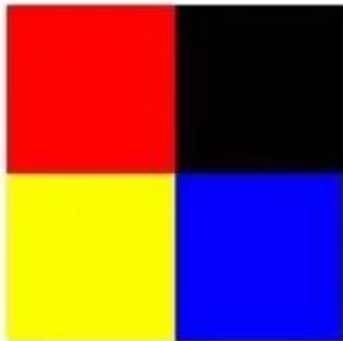
Original Image



11111111	00000000
00000000	00000000
00000000	00000000
11111111	00000000
11111111	00000000
00000000	11111111

Least Significant Bit Steganography

Stego Image



111111 01	000000 11
000000 10	000000 01
000000 00	000000 10
111111 00	000000 11
111111 01	000000 01
000000 01	111111 00



c	a	t
01 10 00 11	01 10 00 01	01 11 01 00

Steganalysis

- Steganalysis is the **art of discovering and rendering** useless such covert messages.
- Steganalysis involves **analysis of the carrier file**
- Simpler steganographic techniques produce some discernible change in the file size, statistics or both.
- These changes can **manifest themselves in color variations, loss of resolution and other distortions** that are visible to the human eye.
- This form of detection **requires that you know what the original carrier image** or file should look like.

- `b_message = ''.join([format(ord(i), "08b") for i in message])`

